Lesson 10.3 - One Tailed and Two Tailed Tests

> **Video:** One Tailed and Two Tailed Tests, Critical Values, & Significance Level by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

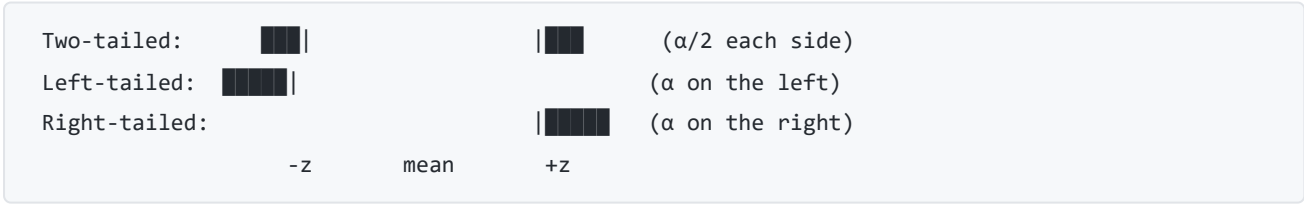
Vocabulary

Term	Meaning
Rejection region	The shaded tail area(s). If the test statistic lands here, reject H_0
Fail-to-reject region	The unshaded middle. If the statistic lands here, keep H_0
Critical value(s)	The z (or t) value(s) separating the two regions
Confidence level c	The area of the fail-to-reject region
Significance level α	The total area of the rejection region: $\alpha = 1 - c$
Calculated value z_c	The **test statistic** computed from the sample (see Lesson 10.4)

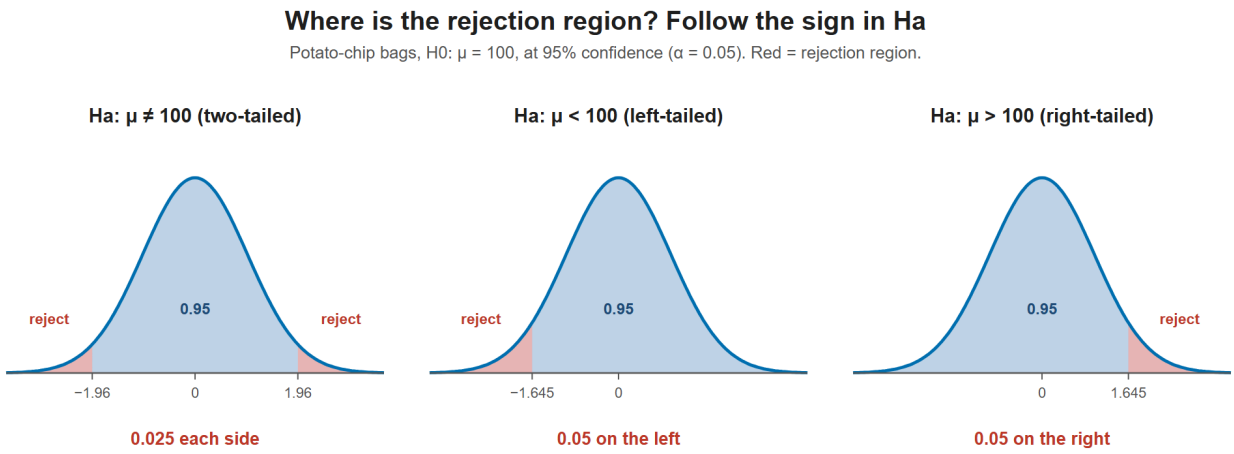
Decision: if z_c falls in the rejection region (beyond the critical value), **reject** H_0. Otherwise, **fail to reject** H_0.

Which test? Read the sign in H_a

H_a	Test	Rejection region	At 95% ($\alpha = 0.05$)
$\mu \neq \mu_0$	**Two-tailed**	Both tails, $\alpha/2$ each	0.025 in each tail, critical $z = \pm 1.96$
$\mu < \mu_0$	**Left-tailed**	Left tail only, area α	0.05 on the left, critical $z = -1.645$
$\mu > \mu_0$	**Right-tailed**	Right tail only, area α	0.05 on the right, critical $z = +1.645$



The **arrow** in H_a points to the tail: $<$ → left, $>$ → right, $\neq$ → both.



The sign in Ha decides where α goes: split between both tails for ≠, all in the left tail for <, and all in the right tail for >.

Example - potato-chip bags

A company says its chip bags average **100 g**.

- An employee thinks the mean is **not** 100 g: $H_0 : \mu = 100$, $H_a : \mu \neq 100$ → **two-tailed**. At 95% confidence, $\alpha = 0.05$ is split into 0.025 per tail, and 0.95 is in the middle.
- If instead $H_a : \mu < 100$ → **left-tailed**, with all 0.05 in the left tail.
- If instead $H_a : \mu > 100$ → **right-tailed**, with all 0.05 in the right tail.

Key takeaways

- $\alpha = 1 - c$ is the significance level, the size of the rejection region.
- $\neq$ → two tails (split α). $<$ → left tail. $>$ → right tail.
- Compare the calculated statistic with the critical value(s) to decide.